

Gelucire 48/16®: A Non-Ionic Solid Surfactant for Solubility Enhancement of Poorly Soluble Drugs

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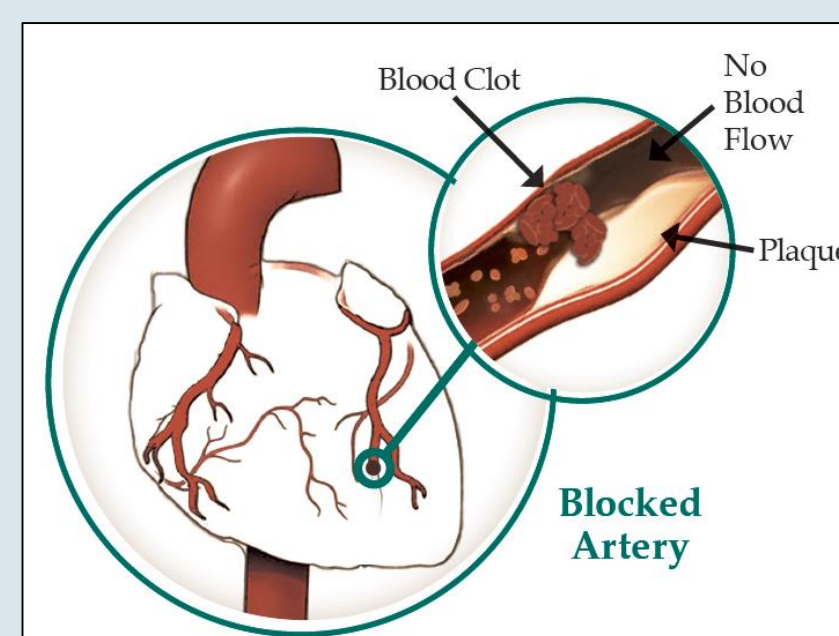


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PURPOSE:

- Ticagrelor (TGR) is an orally active antiplatelet agent, indicated to reduce the rate of thrombotic cardiovascular events.
- A BCS Class-IV drug attributing to its poor solubility and permeability.
- Gelucire 48/16® was evaluated to improve solubility and bioavailability of TGR.
- It is a PEG 32 mono and diesters of stearic and palmitic acid, with an HLB of 16 and CMC of 153±31 mg/L which helps to increase solubility of hydrophobic and lipophobic molecules.
- Being a solid at ambient temperature with a melting point of 48°C, makes it amenable for capsule filling, melt processes and tableting.



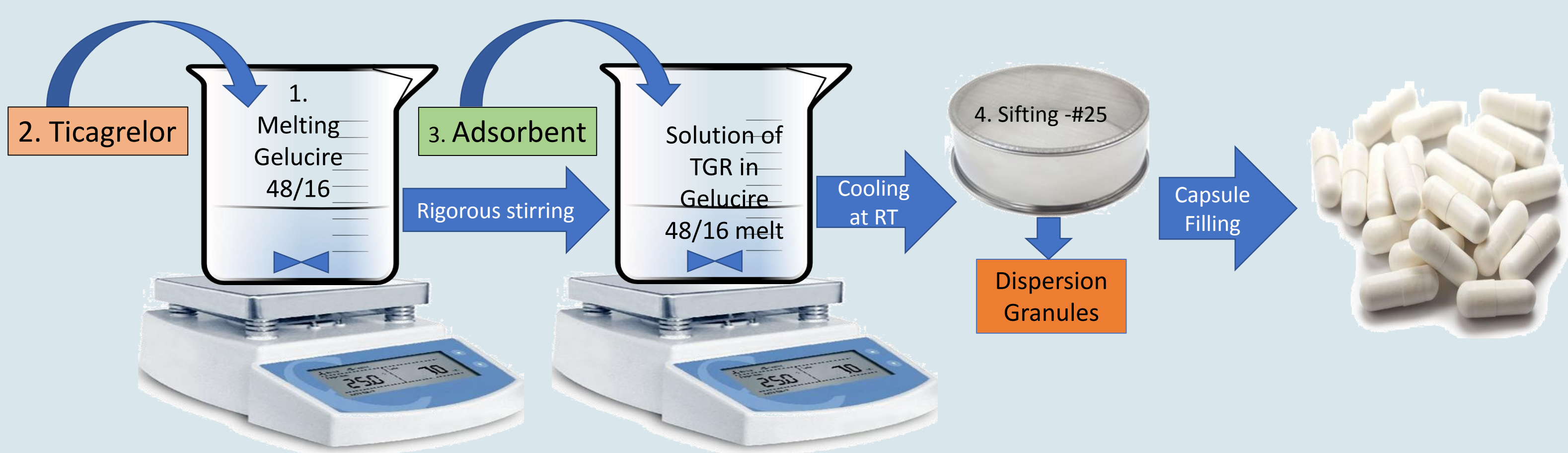
OBJECTIVES:

- To prepare solid dispersions of TGR with Gelucire 48/16® by hot melt technique to improve solubility of TGR.
- To convert the solid dispersion of TGR and Gelucire 48/16® in a oral solid dosage form; from a supgeneric formulation perspective.

METHOD:

Solubility Studies: Solubility of TGR was evaluated in water along with its dispersions with Gelucire 48/16 in the ratios of 1:0.5, 1:1, 1:2, 1:3, 1:4 and 1:5. The samples were stirred for 72h at 37±0.2°C and evaluated for drug content in supernatant after centrifugation. Optimum solubility was observed with 1:3 ratio of TGR: G-48/16 and therefore this ratio was further evaluated.

Formulation: Solid dispersions were prepared by hot melt technique to improve solubility of TGR.



Step-1: Gelucire 48/16 was melted at 70°C

Step-2: TGR sieved through #40 and added in the molten mass of Gelucire 48/16 under constant stirring-TGR solubilizes

Step-3: Solution of TGR adsorbed on a suitable adsorbent i.e. Avicel PH200, Prosolv SMCC90, Pearlitol, Syloid XDP3150, Aeroperl 300 etc.

Step-4: Cooling, and sifting through #25 mesh followed by capsule filling.

Prosolv SMCC90 was found to be a better adsorbent in terms of dissolution rate compared to other adsorbents evaluated, therefore further studies were carried out with it.

Differential Scanning Calorimetry: Samples (5-10 mg) were sealed in aluminium pans for analysis. The DSC thermograms were recorded from 30°C to 300°C at a heating rate of 10°C/min. A nitrogen flow rate of 20 ml/min was used for each DSC run.

X-Ray Diffraction Study: Powder X-ray diffraction patterns of samples were recorded to assess the solid state of TGR, using a Phillips X-ray diffractometer with a copper target, voltage 40 kV, current 20 mA, at a scanning speed of 2° per min.

Fourier Transform Infrared Spectroscopy (FTIR) : The KBr disks of samples were prepared using electrically operated KBr Press. A PerkinElmer Fourier transform spectrophotometer was used to record IR spectra of the prepared discs, to confirm for any interaction of TGR with excipients of dispersion granules.

Capsule Preparation: The hot melt was directly filled into hard gelatin capsules as well as adsorbed on suitable inert carriers to obtain granules, which were filled in hard gelatin capsules.

In-Vitro Evaluation of Capsules: The capsules were evaluated for disintegration time and *in vitro* dissolution in 900 mL distilled water using USP Type-II apparatus with sinker, at 75 rpm and 37.5±0.2°C. Dissolution of branded tablets was also performed. Analysis of the samples was performed on a UV spectrophotometer at λmax 220 nm after suitable dilutions.

RESULTS:

Solubility Studies:

- The results of solubility study of TGR shown in Figure-1.
- Solubility of TGR (~9 µg/mL) was found to increase with an increasing concentration of Gelucire 48/16.
- Saturation solubility was achieved at 1:3 ratio (TGR:Gelucire 48/16).

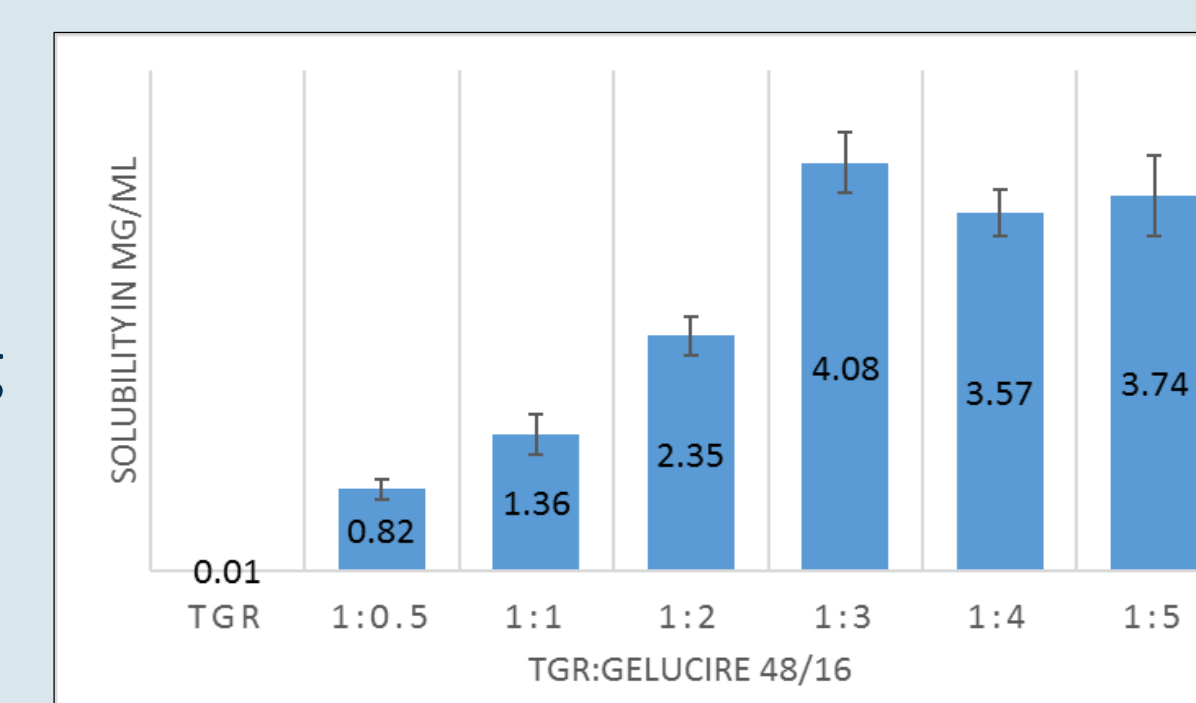


Figure-1: Solubility Study of TGR in water.

Differential Scanning Calorimetry:

- DSC thermograms (Figure-2) indicates sharp endotherms of TGR and Gelucire 48/16 at 140.82°C and 50.02°C respectively.
- TGR is solubilized in the melt of Gelucire 48/16 which could explain the disappearance of TGR endotherm.
- Endotherm of TGR was not evident in the solid dispersions indicating a change in the physical state, attributing to its enhanced solubility.

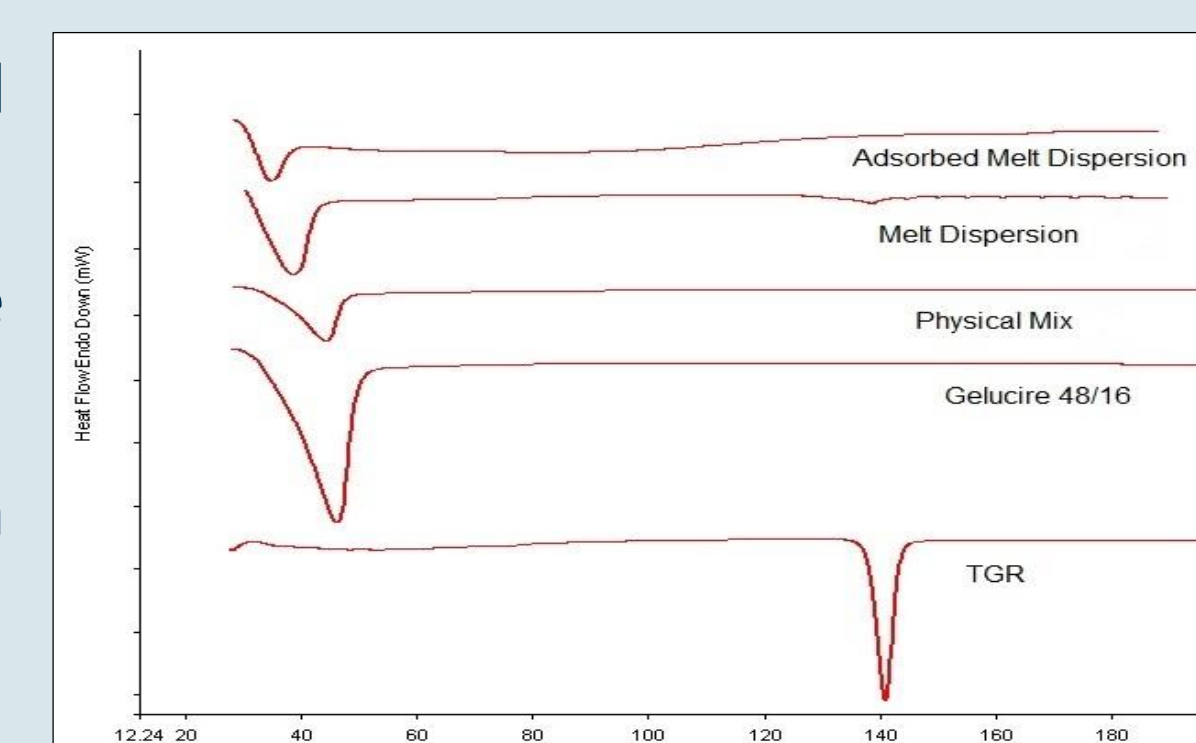


Figure-2: DSC Thermograms .

X-Ray Diffraction Study:

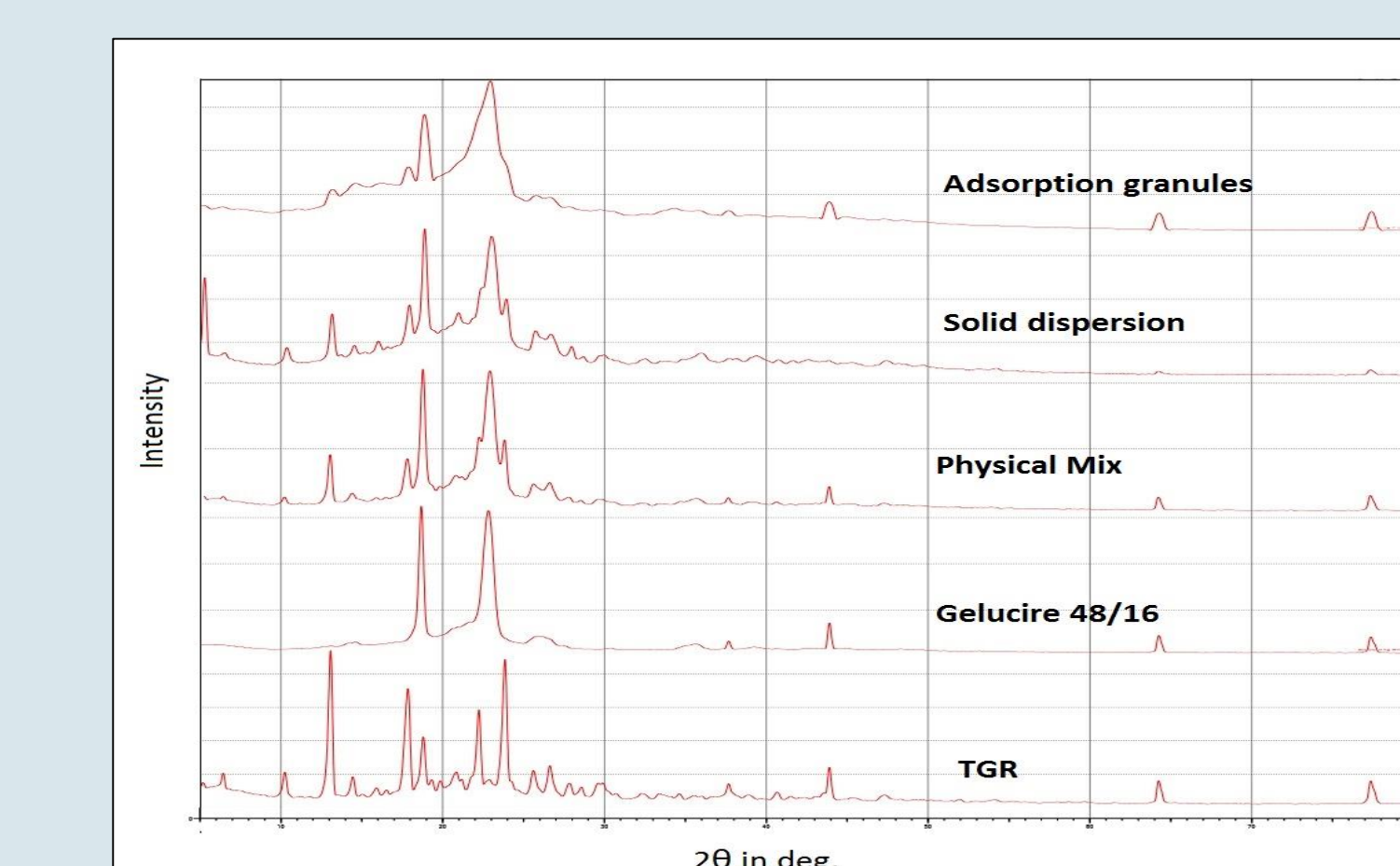


Figure-3: X ray diffractograms.

- XRD pattern (Figure-3) of TGR indicates its crystalline nature
- Gelucire-48/16 shows a semi-crystalline nature
- Physical mix shows the merging of peaks of drug with the peaks of Gelucire 48/16.
- Solid dispersion indicates change in the crystalline form of TGR in the matrix of Gelucire-48/16.
- Presence of adsorbent may prevent the reversion of the amorphous drug to crystalline state on storage of the solid dispersion.

Fourier Transform Infrared Spectroscopy (FTIR) :

- FTIR spectra depicted in Figure-4 shows- TGR has several characteristic peaks at 670, 1330, 1385, 1521, 1588 cm⁻¹.
- These peaks are of diagnostic value to elucidate drug-excipient interactions
- These peaks are also visible in the spectra of physical mix, solid dispersion and dispersion granules
- FTIR spectrum of dispersions did not reveal any significant drug excipient interaction

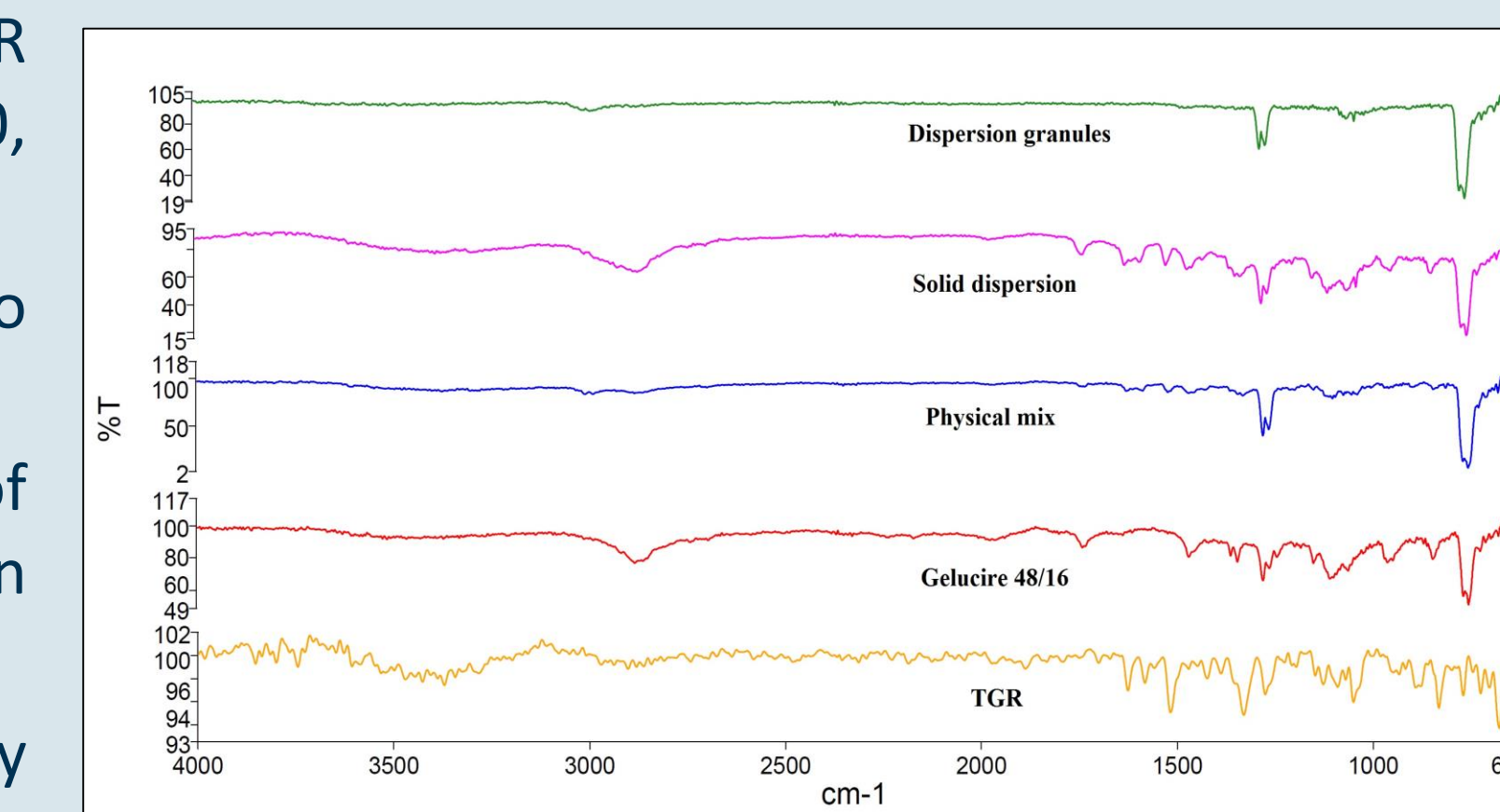


Figure-4: FTIR Spectra.

In-Vitro Evaluation of Capsules:

- Disintegration time of capsules was found to be 8-10 min.
- In vitro* dissolution of in house formulation and branded tablets is shown in Figure-5.
- Dissolution profile of in house capsules of TGR showed remarkably improved dissolution rate in distilled water compared to branded tablets.
- Capsules with melt dispersion showed ~90% drug release in 10 min, capsules of dispersion granules showed >80% release in 30 min.

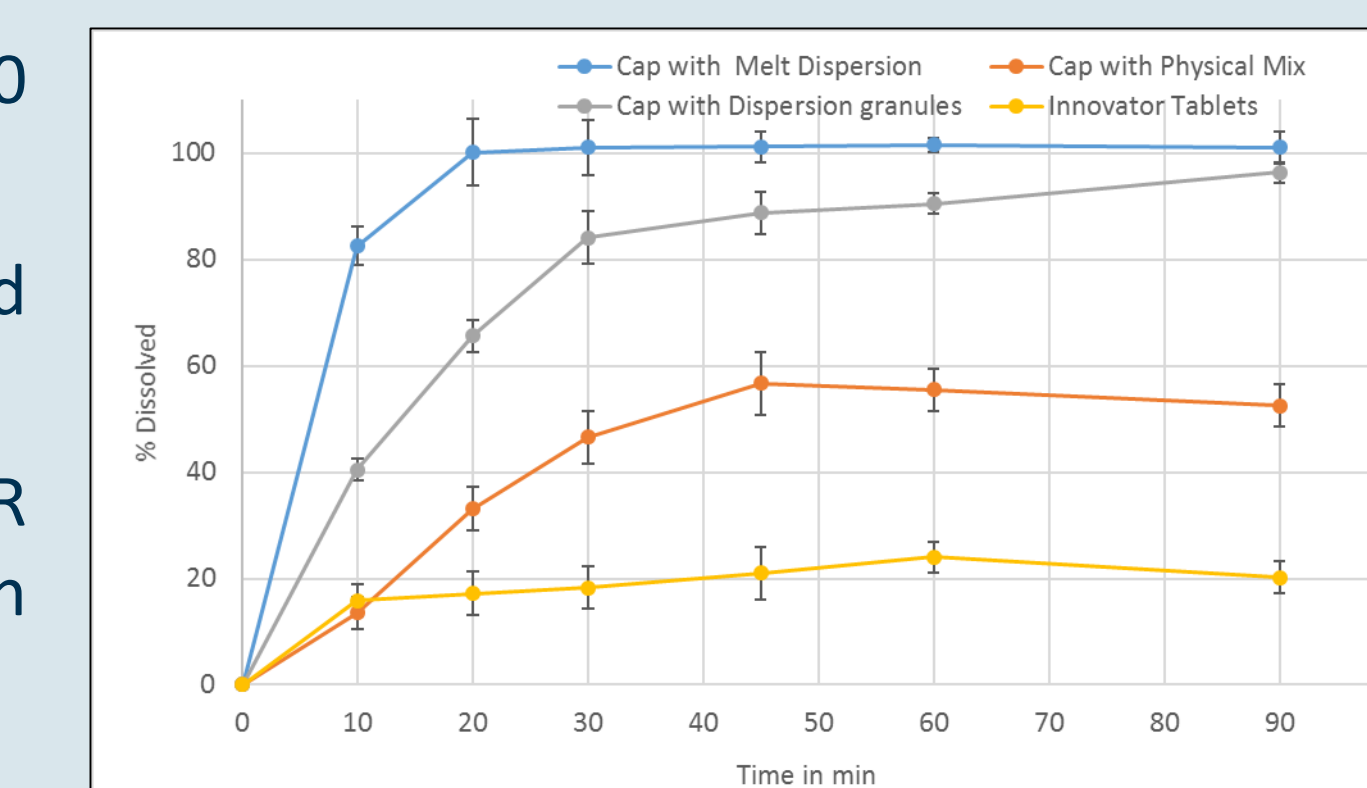


Figure-5: In vitro dissolution test.

- Branded tablets showed ~20% drug release over a period of 90 min.

CONCLUSIONS:

- Gelucire 48/16 was employed effectively to enhance the solubility and dissolution of a poorly water-soluble BCS Class-IV drug i.e. TGR.
- DSC and XRD studies indicated a change in physical state of TGR in Gelucire 48/16 matrix, which could be attributing to its enhanced solubility and dissolution rate.
- FTIR analysis did not indicate any significant interaction between TGR and excipients.
- Gelucire 48/16 is thus an effective solubilizing excipient for solubility and dissolution enhancement of poorly soluble molecules.
- The formulated capsules of TGR thus offer a formulation with supgeneric potential.
- In vitro* lipidolysis and animals studies are planned to further confirm the potential of Gelucire 48/16 as an effective solubilizer and bioavailability enhancer for TGR.

REFERENCES:

- A. R. Shrivastava, B. Ursekar and C. J. Kapadia, Cur. Drug Del., 2009, 6, 28-37.
- Serajuddin, A.T.M. J. Pharm. Sci., 1999, 88, 1058-66.
- https://www.accessdata.fda.gov/drugsatfda_docs/label/2015/022433s015lbl.pdf
- www.gattefosse.com

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